

Modification of SIWRS Model for Cholera Transmission to Include Vaccination Compartments

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Abstract

Since the early 19th century, cholera, a water-borne bacterial infection caused by the ingestion of the bacterium *Vibrio Cholerae*, has been a highly prevalent disease in many areas throughout the world. These areas often have limited access to professional healthcare, and vaccinations for endemic diseases, including cholera, are far too expensive and inaccessible for the general population, thus creating scenarios where mass immunization is not considered a realistic priority. This study focuses on the diffusion of cholera in a dynamic population while accounting for various vaccination compartments in order to predict cholera prevalence for a potential present-day epidemic. The original system, a SIWRS model which analyzes the data set of the 2006 cholera epidemic in Angola, lacks a vaccination compartment. The modified systems use parameters from the same dataset but also incorporate individuals vaccinated with perfect and imperfect vaccines into the models. Graphical and stability analysis find stable equilibrium's for the coexistence of individuals over time and within fluctuating vaccination rates. Understanding the spread of disease is crucial to the mitigation of damage to a population in the event of a present-day epidemic.

1 Introduction

Cholera is a water-borne, bacterial infection caused by ingestion of the bacterium *Vibrio Cholerae*. The bacterium is typically found and transmitted through water and food that has been contaminated by a vector already infected with *V. Cholerae*. Its symptoms include severe diarrhea and dehydration, both of which can lead to an extreme, rapid loss of bodily fluids. Without treatment, an infected patient may die within a matter of hours. Out of the estimated 1.3–4 million cases of cholera annually, between 21,000 and 43,000 people die from it each year [1].

The original reservoir of *V. Cholerae* is located in the Ganges delta in India. Around the 19th century, cholera began to spread from this reservoir to many other parts of the world, subsequently causing six devastating pandemics throughout the globe. A seventh pandemic began in South Asia in 1961 and spread through Africa and the Americas, leaving cholera as an endemic disease in many countries throughout the world [3]. Among these countries was Angola, a southwestern African country with a population of 34.5 million as of 2021. From February 2006 to September 2007, Angola suffered one of the worst cholera outbreaks in its history, with around 83,520 total cases and 3,140 deaths during this period. The outbreaks were reported as being caused by poor water quality and a lack of accessible sanitation services [4].

The first cholera vaccine was developed in the late 1800s by Spanish physician Jaume Ferran. Ferran used live cultures from cholera patients from Marseilles, France [11]. This vaccine was later used on over 30,000 people in Spain during the cholera epidemic in 1884. In 2016, Vaxchora, or CVD 103-HgR, became the first cholera vaccine to be approved by the Food and Drug Administration (FDA) in the United States. As of 2023, it remains the only cholera vaccine available in the United States. Vaxchora is a two-dose oral vaccine that is recommended by the FDA when traveling to a country where cholera is recurrent [12].

In 2017, researchers published a paper entitled “Model distinguishability and inference robustness in mechanisms of cholera transmission and loss of immunity”, in which they used the dataset from the 2006 Angola epidemic to evaluate existing SIWR models for cholera and added necessary modifications [10]. However, while these models accurately represent the 2006 epidemic, they could not correctly predict the equilibrium points and steady states of a present-day epidemic because the vaccine in the study was approved in 2016, thus there was not enough data gathered. The primarily used vaccine, Vaxchora, has a 59% efficacy rate, thus classifying it as an imperfect vaccine [14]. The Exponential model published by Lee et al. is modified to more accurately model a potential present-day epidemic by including a vaccination compartment in the model. The vaccination compartment for this model is based on the vaccination compartment in an SVIR model for COVID-19 published in the paper entitled “Modelling the impact of perfect and imperfect vaccination strategy against SARS CoV-2 by assuming varied vaccine efficacy over India”. They include models with both perfect and imperfect vaccination compartments [16]. Similarly, this paper aims to use the modified SIWRS model to compare the spread of cholera in a theoretical population vaccinated with a perfect vaccine, a population vaccinated with an imperfect vaccine, and a population left unvaccinated.

2 Mathematical Model

The SIWRS (Susceptible, Infected, Pathogen Concentration in an Aquatic Reservoir, Recovered) Model of Tien and Earn's paper on multiple transmission pathways and disease dynamics in a water-borne pathogen [15] model has been modified to better fit the provided data analysis for the inclusion of vaccination compartments. The overall study will contain three models of different theoretical scenarios in a dynamic population: One model with no vaccines available or existent, another with a perfect vaccine that guarantees immunity after vaccination, and finally one with a leaky, imperfect vaccine that has a non-100% efficacy, which is a more realistic application to the real world. All three models are tested with the dataset from the 2006 Angola epidemic to check for consistency in the stable equilibrium points. These models also considers numerous factors that affect a population, including the force of infection from person-to-person contact and water transmission pathways, the birth rate, the death rate for each subsection of the population, the decay rate of cholera in water, the recovery rate, and the loss of immunity.

2.1 Assumptions

Similar to a SIR model, it is assumed that there is a homogeneous mixing of the infected and susceptible populations and that the natural birth and death rates are constant throughout the model. In addition, the death rates for each subgroup of susceptible, infected, and recovered individuals are the same, so they are negatively proportional to each respective group. It is also assumed that individuals who are infected cannot die from cholera. Furthermore, to simplify the model, the vaccination rate is constant throughout the time period of the system, as defined by v . Since there are many environmental factors that affect the rate of transmission, β_I and β_W are generalizations of short- and long-term transmission, respectively, with β_I representing food-borne, household, and non-environmental factors and β_W representing bacterial contamination in water, which could be affected by the variations in temperature, precipitation, and humidity[15], [13]. In the perfect vaccine model, the assumption is made that after a person has been vaccinated, they are unconditionally immune, meaning they can't be infected. However, this being said, in this model, if a person recovers from cholera and is not vaccinated, they are still susceptible to being infected again. For both vaccinated models, $S + I + R + V = 1$, and all populations have an equal death rate. In the imperfect vaccine model, the efficacy of the vaccination is assumed to remain constant throughout the time period.

2.2 Initial Equations

Below is a flow diagram of a classic SIWRS model and a Exponential model for cholera transmission[10], [8]:

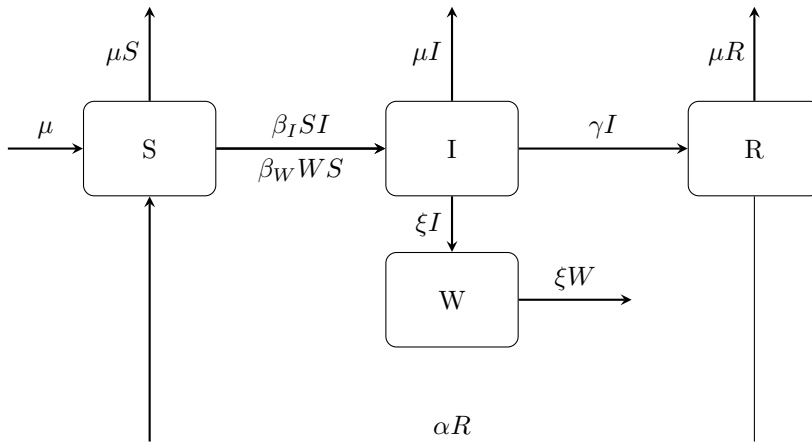


Figure 1: Flow Diagram of SIWRS Model

$$\begin{aligned}
\frac{dS}{dt} &= \mu - \beta_I SI - \beta_W WS - \mu S + \alpha R \\
\frac{dI}{dt} &= \beta_W WS + \beta_I SI - \gamma I - \mu I \\
\frac{dW}{dt} &= \xi(I - W) \\
\frac{dR}{dt} &= \gamma I - \mu R - \alpha R
\end{aligned} \tag{1}$$

The above equations are the original SIWRS model developed for the 2006 Angola cholera epidemic. The first equation, $\frac{dS}{dt}$, refers to the rate of change in the size of the susceptible population. It is affected by factors such as the natural birth and death rates of the population, transmission rates of disease, pathogen concentration, and loss of immunity in recovered individuals. The next equation, $\frac{dI}{dt}$, refers to the rate of change in the size of the infected population. Individuals enter the infected population from the susceptible population and leave either by entering the recovered population or dying from disease. Next, $\frac{dW}{dt}$ refers to the rate of change of pathogen concentration in water and is affected by the size of the infected population and the decay rate of cholera in water. Lastly, $\frac{dR}{dt}$ is the rate of change of the recovered population. Individuals enter the recovered population from the infected population and leave either to the susceptible population through loss of immunity or by dying.

2.3 Modifications to Initial Equations

2.3.1 Perfect Vaccine Compartment:

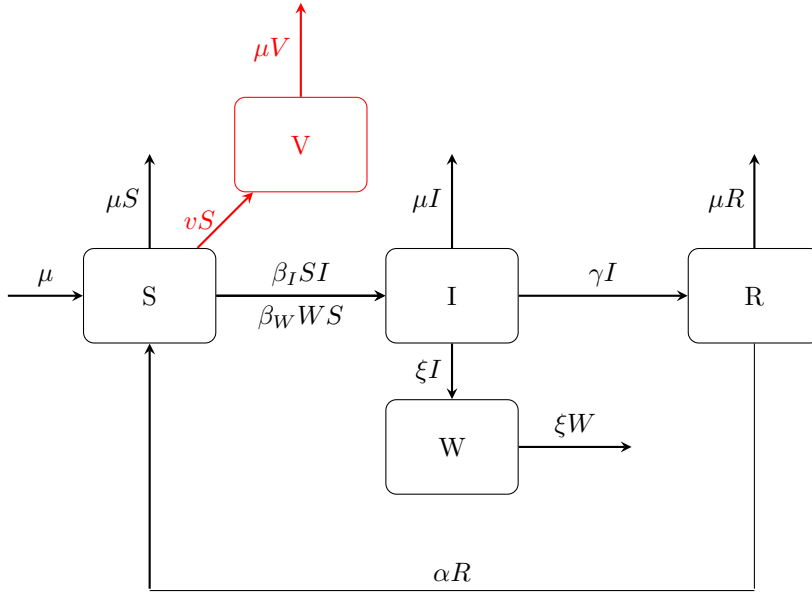


Figure 2: Flow Diagram of SIWRS Model with Perfect Vaccine Compartment

Red denotes the changes of the new model from the previous model

$$\begin{aligned}
\frac{dS}{dt} &= \mu - \beta_I SI - \beta_W WS - \mu S + \alpha R \\
\frac{dI}{dt} &= \beta_W WS + \beta_I SI - \gamma I - \mu I \\
\frac{dW}{dt} &= \xi(I - W) \\
\frac{dR}{dt} &= \gamma I - \mu R - \alpha R \\
\frac{dV}{dt} &= vS - \mu V
\end{aligned} \tag{2}$$

The above equations are consistent with the original SIWRS model, modified with a perfect vaccination compartment. When creating the perfect vaccination compartment model, the assumption is made that once a person is vaccinated, they are perfectly immune and cannot be either infected or return to the susceptible population. The addition of this compartment, $\frac{dV}{dt}$, being the rate of vaccinated individuals, is proportional to the susceptible population by the vaccination rate of cholera (vS), which likewise is subtracted from the susceptible population.

2.3.2 Imperfect Vaccine Compartment:

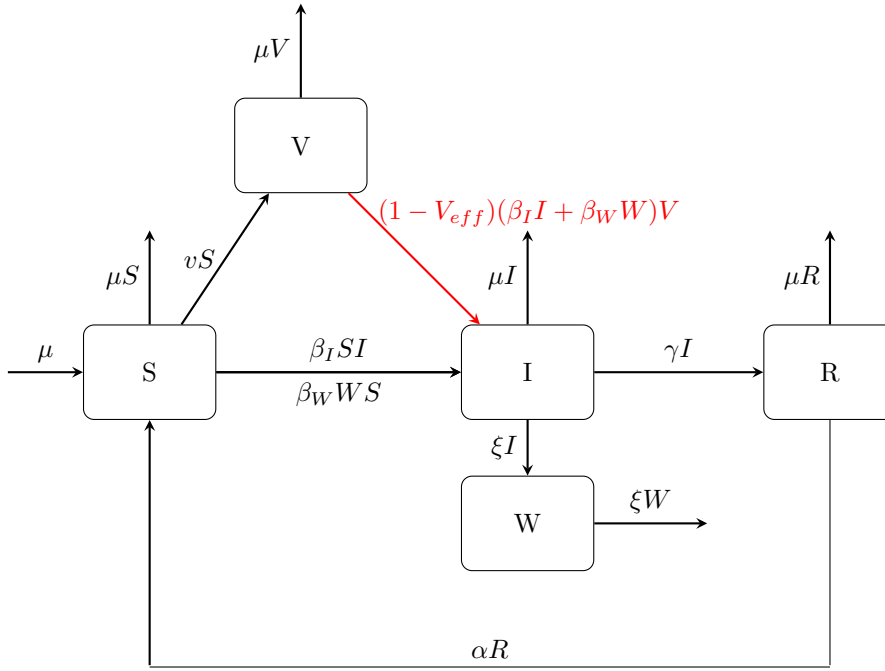


Figure 3: Flow Diagram of SIWRS Model with Imperfect Vaccine Compartment

$$\begin{aligned}
\frac{dS}{dt} &= \mu - \beta_I SI - \beta_W WS - \mu S + \alpha R \\
\frac{dI}{dt} &= \beta_W WS + \beta_I SI - \gamma I - \mu I + (1 - V_{eff})(\beta_I I + \beta_W W)V \\
\frac{dW}{dt} &= \xi(I - W) \\
\frac{dR}{dt} &= \gamma I - \mu R - \alpha R \\
\frac{dV}{dt} &= vS - (1 - V_{eff})(\beta_I I + \beta_W W)V - \mu V
\end{aligned} \tag{3}$$

The above equations are the original SIWRS model modified with an imperfect vaccination compartment. Similarly, this model has the vS term from the perfect vaccination model, which accounts for the number of vaccinated individuals $\frac{dV}{dt}$. However, it also includes the term $(1 - V_{eff})(\beta_I I + \beta_W W)V$ for the amount/percentage of people in whom the vaccine is not effective, also known as the force of infection. This term is subtracted from the vaccinated population because immunity is not guaranteed for them, and added to the infected population due the ability to be reinfected after vaccination and imperfect efficacy. In the original SIRV model, $\beta_I I + \beta_W W$ is defined by λ , where λ was defined as $\beta * I(t)$. However, for a SIWRS model, two rates of transmission are present: β_I (Person-to-person infection rate) and β_W (Water-Borne Transmission Rate). Therefore, we equate λ_{tot} to be the summation $\beta_I I + \beta_W W$ (See Table 2.4).

2.4 Parameters

List of Parameters			
Parameter	Description	Value	Units
μ	Natural Birth and Death Rate in Angola	0.000557	$days^{-1}$
γ	1/(Mean Duration of Infectiousness), or Recovery Rate	1/4	$days^{-1}$
β_I	Person-Person infection rate	0.264	$days^{-1}$
β_W	Water-Borne Transmission Rate	1.21	$days^{-1}$
α	Loss of Immunity Rate	$\frac{1}{365}$	$days^{-1}$
ϵ	Decay Rate of Cholera in Water	0.00756	$days^{-1}$
V_{eff}	Vaccine Efficacy	0.59	unitless
v	Vaccination Rate	non-constant	$\frac{Vaccinated(V)}{day}$

Definitions and Numerical Values for Variables [7], [10]

Populations	
Population	Description
S	Susceptible Individuals
I	Infected Individuals
W	Pathogen Concentration in an Aquatic Reservoir
R	Recovered Individuals
V	Vaccinated Individuals

3 Methodology

3.1 Finding Eigenvalues and Stability

3.1.1 No Vaccine Model

In addition to graphical analysis, finding stability shows the resulting distribution of the population. In order to find the critical points, each differential equation is set to zero. Setting the initial conditions, the software Maple calculates the critical points, which for the no vaccination model, is (S=1, H=0, W=0, R=0) and (S=0.1700, I=0.01080, W=0.01080, R=0.8192).

After solving for the critical points, determining the steady states for the equilibriums help show the coexistence of the population and the theoretical situation for each model. A Jacobian matrix is used to determine the stability for each critical point, which follows the simplified form of:

$$\frac{dx}{dt} = f(x, y) \quad \frac{dy}{dt} = g(x, y)$$

$$J = \begin{bmatrix} \frac{\partial F}{\partial x} & \frac{\partial F}{\partial y} \\ \frac{\partial G}{\partial x} & \frac{\partial G}{\partial y} \end{bmatrix}$$

Finding the partial derivatives for each function of the original model (eq. 1), the Jacobian matrix is:

$$J = \begin{bmatrix} -0.264H - 1.21W - 0.000557 & -0.264S & -1.21S & 1/365 \\ 0.264H + 1.21W & 0.264S - 0.2506 & 1.21S & 0 \\ 0 & 0.00756 & -0.00756 & 0 \\ 0 & 0.25 & 0 & -0.003297 \end{bmatrix}$$

With each point (S=1, H=0, W=0, R=0) and (S=0.1700, H = 0.01080, W = 0.01080, R=0.8192) having its own Jacobian, substituting each point gives the following matrices:

$$J_1 = \begin{bmatrix} -0.000557 & -0.264 & -1.21 & 1/365 \\ 0 & 0.01344 & 1.21 & 0 \\ 0 & 0.00756 & -0.00756 & 0 \\ 0 & 0.25 & 0 & -0.003297 \end{bmatrix}$$

and,

$$J_2 = \begin{bmatrix} -0.003277 & -0.04488 & -0.2057 & 1/365 \\ 0.002720 & -0.2057 & 0.2057 & 0 \\ 0 & 0.00756 & -0.00756 & 0 \\ 0 & 0.25 & 0 & -0.003297 \end{bmatrix}$$

To conclude whether a critical point is asymptotically stable, all eigenvalues' real parts must be negative, or the model fails to reach a steady equilibrium point and approaches infinity. To calculate the eigenvalues for any matrix J , λ is an eigenvalue of J if $\det(A - \lambda I) = 0$, where $\det()$ is the determinant of the matrix and I is the identity matrix. Solving for the eigenvalues of J_1 and J_2 :

$$\lambda_1 = \begin{bmatrix} -0.000557 \\ 0.09916 \\ -0.09328 \\ -0.003297 \end{bmatrix}$$

and,

$$\lambda_2 = \begin{bmatrix} -0.2127 \\ -0.000557 \\ -0.003262 + 0.003720i \\ -0.003262 - 0.003720i \end{bmatrix}$$

Since there is a positive real part in one of the eigenvalues for J_1 , the point (S=1, H=0, W=0, R=0) is unstable, meaning that the model will not reach a finite equilibrium over an infinite time with these conditions. On the other hand, the eigenvalues of J_2 are all negative and therefore the point (S=0.1700, H = 0.01080, W = 0.01080, R=0.8192) is asymptotically stable.

3.1.2 Perfect and Imperfect Vaccine Model

Repeating the same process for the perfect and imperfect vaccination models, the stable critical points and eigenvalues after solving its Jacobian matrices are (eq. 2, eq. 3):

$$(S = 0.02710, H = 0, W = 0, R = 0, V = 0.9729) \\ \text{(Perfect Vaccine)}$$

$$\lambda = \begin{bmatrix} -0.000557 \\ -0.001057 \\ -0.2445 \\ -0.006514 \\ -0.003297 \end{bmatrix}$$

and for the imperfect vaccination model,

$$(S = 0.07163, H = 0.008961, W = 0.008961, R = 0.6795, V = 0.2399) \\ \text{(Imperfect Vaccine)}$$

$$\lambda = \begin{bmatrix} -0.2123 \\ -0.02268 \\ -0.0005570 \\ -0.002011 + 0.001289i \\ -0.002011 + 0.001289i \end{bmatrix}$$

3.2 Stability Analysis

For the original no-vaccine model, 81.9% are recovered and 17% are susceptible, with the rest being the infected population. The stability point for the perfect model is close to the extinction of cholera, with the vaccinated individuals V in the compartment around 97.29% and the susceptible being 2.71% of the total population. This is because over an infinite time t , the recovered individuals lose immunity by a rate of $\alpha = \frac{1}{365}$ and become susceptible, and so they are vaccinated by the vaccination rate v , meaning almost all individuals will become vaccinated.

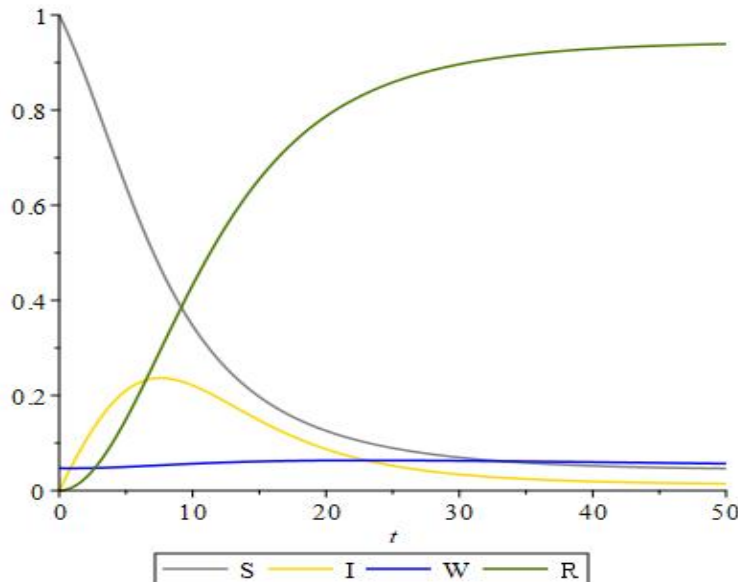
Conversely, the imperfect vaccine establishes the coexistence of all individuals and water pathogens. Due to the vaccine efficacy within the term $(1 - V_e f f)(\beta_I I + \beta_W W)V$, those who are vaccinated are not guaranteed immunity, so the infected population is able to become stable at $I \neq 0$. This is the more realistic situation because, with a non-100% efficacy vaccine, no population is able to be the sole group of individuals, which can be compared to the current COVID-19 epidemic; as of July 2023, 31% of the USA population is recovered, 31.6% infected, the rest is the susceptible population, and 81.4% has at least the first dose [2]. Although the data analysis shows a stability point for vaccinated individuals at 24%, this is due to the assumption that vaccinated individuals are a separate group from the SIR distribution. The COVID-19 vaccine efficacy is 75%-86% compared to the 59% for cholera [9], and COVID-19 also includes multiple doses of vaccination, including the first dose, primary series, and bivalent booster dose [1]. The stable equilibrium ($S=0.07163$, $H=0.008961$, $W=0.008961$, $R=0.6795$, $V=0.2399$) is a practical solution to the effects of the Vaxchora vaccine on cholera transmission and can be compared to the current event of COVID-19.

4 Results

In order to observe the effects of vaccination on cholera spread through populations, all three models were graphed over time (in days). The different populations (susceptible, infected, recovered, and vaccinated) were represented as decimal proportions of the total population. Models were evaluated over varying time intervals and with varying vaccination rates.

4.1 Effects of Vaccination over Time

The models were observed at three different time intervals: a 50-day outbreak period, a 100-day outbreak period, and a 1-year outbreak period. At $t=0$ (t is in terms of days), the susceptible population is at 1, because none of the population has been infected or vaccinated yet. The vaccination rate was kept constant at 2% of the population vaccinated per day for both vaccination models. Over time, the models show greater divergence in their effects.



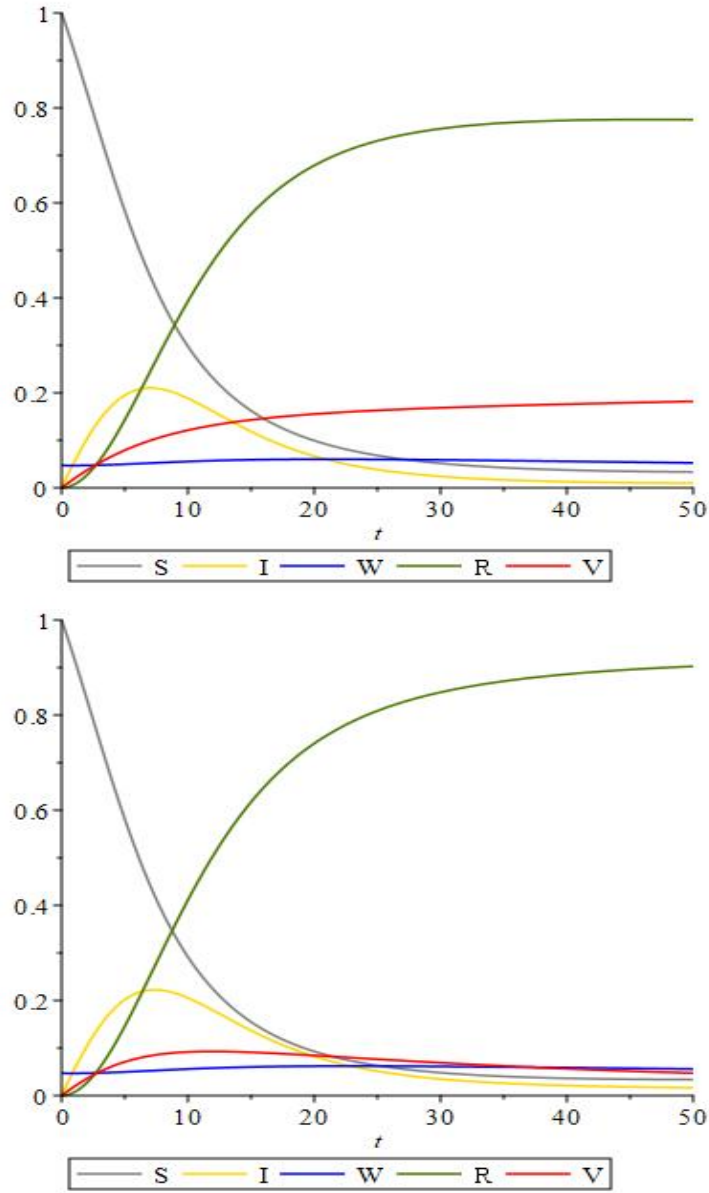


Figure 1: No Vaccine, Perfect Vaccine, and Imperfect Vaccine ($t=50$)

At $t=50$, there is already a considerable difference between the unvaccinated population and the population vaccinated with the perfect vaccine. In the population with the perfect vaccine, there are much smaller infected ($\approx 20.9\%$) and recovered populations and a steady vaccinated population, while the no-vaccine model had the greatest maxima of infected individuals ($\approx 23.6\%$) and the imperfect vaccine as the medium ($\approx 22.2\%$). The infected population also reach the maximum at different time intervals; the no vaccine between 7.44 and 7.63 days, the perfect between 6.93 and 7.11 days, the imperfect between 7.25 and 7.4 days, meaning that the effectiveness of the vaccine correlates to the period of time for the disease to approach its peak. The difference between the unvaccinated population and the population vaccinated with the imperfect vaccine is much smaller, and after a period of 50 days, their susceptible, infected, and recovered populations are relatively the same size.

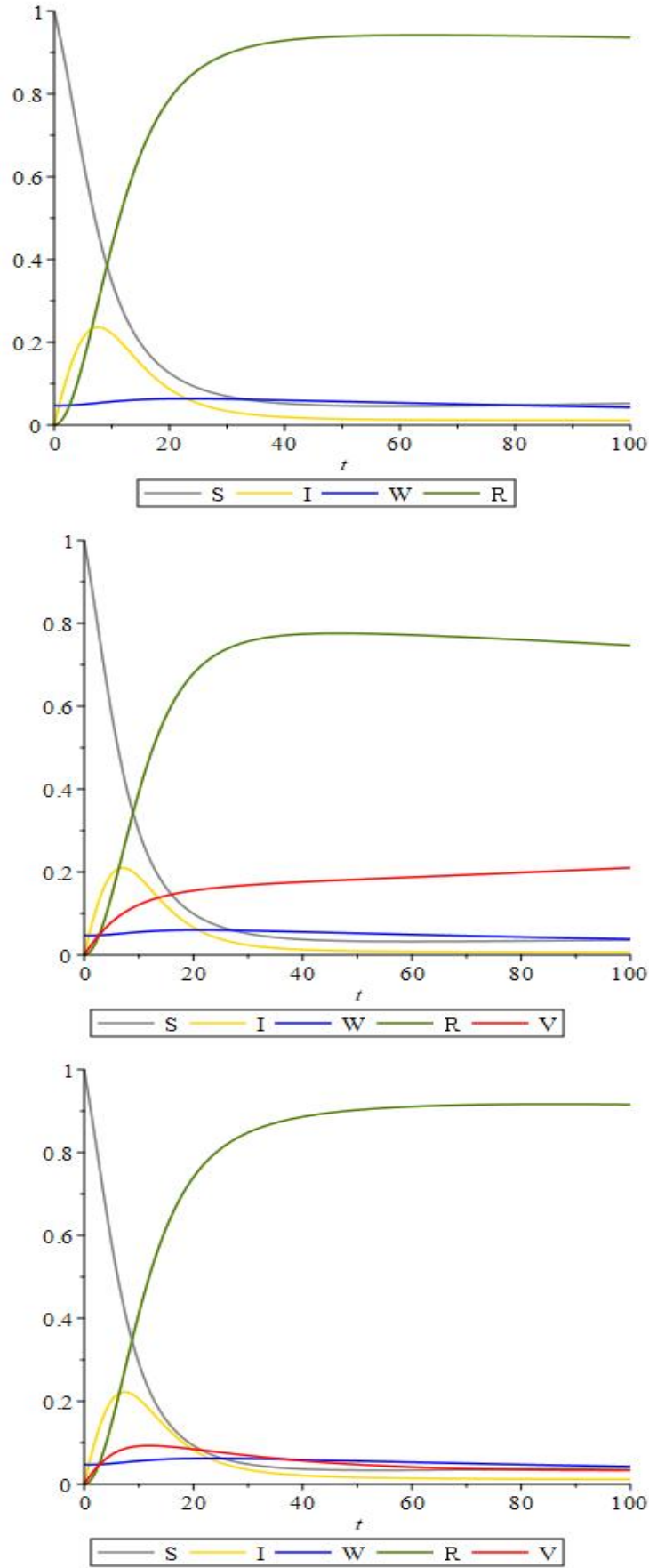
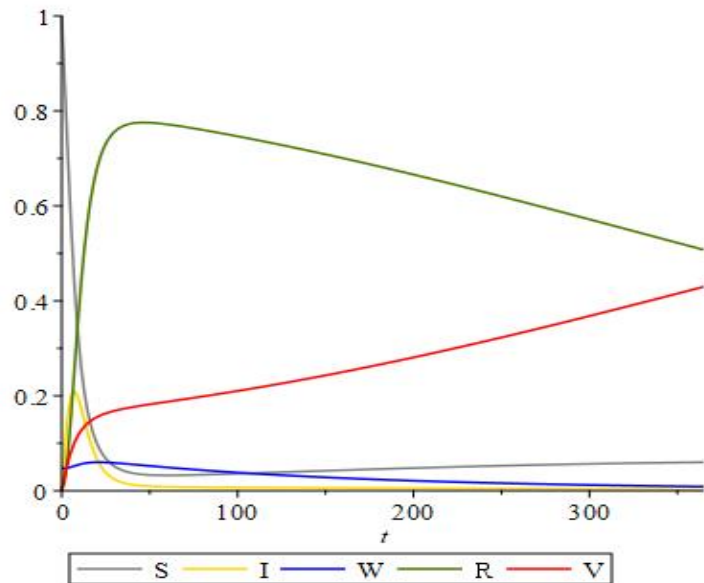
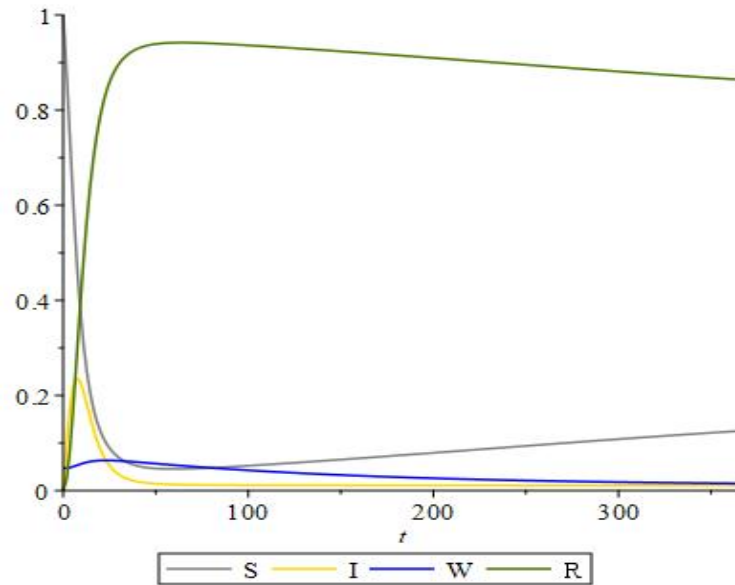


Figure 2: No Vaccine, Perfect Vaccine, and Imperfect Vaccine ($t=100$)

At $t=100$, the unvaccinated population and the population vaccinated with the perfect vaccine stay relatively stable. There is not a considerable change between these graphs and the graphs at $t=50$ for these two populations. However, in the population vaccinated with an imperfect vaccine, there is a steady decrease in the vaccinated population due to loss of immunity and the percentage of vaccinated individuals for whom the vaccine was not effective.



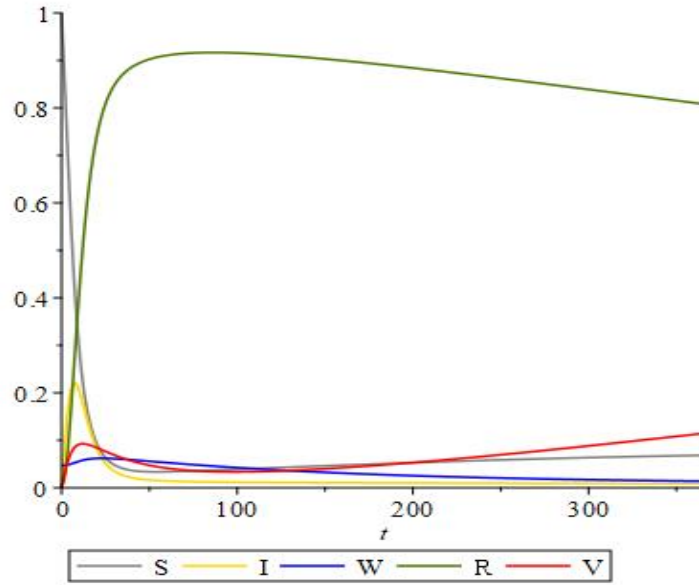


Figure 3: No Vaccine, Perfect Vaccine, and Imperfect Vaccine ($t=365$)

At $t=365$, after a one-year period of the outbreak, the percentage of the unvaccinated population who recovered starts to steadily decline due to loss of immunity, and there is a corresponding increase in the size of the susceptible population. A growing susceptible population creates a higher potential for a second outbreak in the future. In the population vaccinated with the perfect vaccine, the vaccinated proportion of the population consistently increases, and the susceptible population is beginning to stabilize. Lastly, in the population vaccinated with the imperfect vaccine, there was a dip in the vaccinated population around $t=100$ due to loss of immunity. However, after that initial dip, the vaccinated population shows steady growth, the recovered population decreases as individuals move to the vaccinated population, and the susceptible population approaches stabilization. Further graphical analysis indicates that these populations stabilize at around $t=800$, which would be around 2.2 years.

4.2 Effects of Changes in Vaccination Rate

The perfect vaccine and imperfect vaccine models were additionally observed at 3 different vaccination rates: 0.5% of the population vaccinated per day, 1% of the population vaccinated per day, and 2% of the population vaccinated per day. These are theoretical vaccination rates for a country during an epidemic so that the different rates can be compared and show association with the distribution of the population; they are not based on the 2006 Angola dataset. The models were observed at these vaccination rates for a period of 100 days.

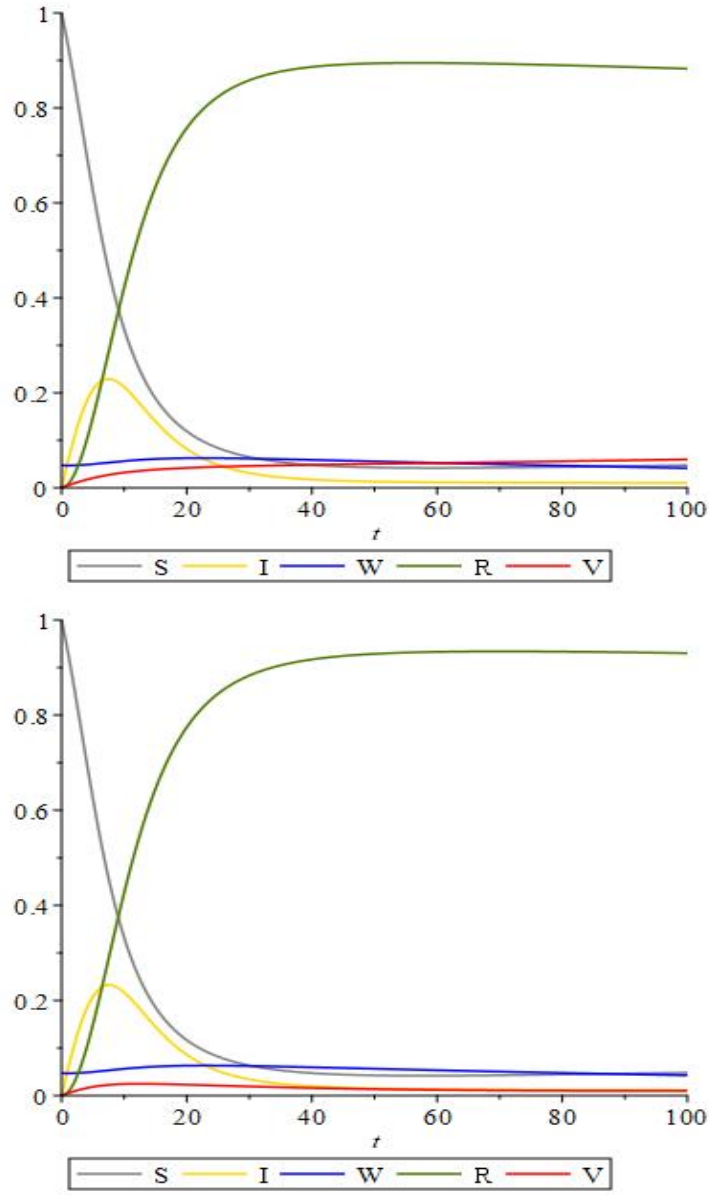


Figure 4: Perfect Vaccine, and Imperfect Vaccine ($v=0.005$)

At **0.5% of the population vaccinated per day**, the proportion of the population vaccinated barely increases for the perfect vaccine and then dips almost to 0 for the imperfect vaccine. The susceptible and recovered populations show little to no change either, which implies that it will take a much longer time period for them to reach equilibrium.

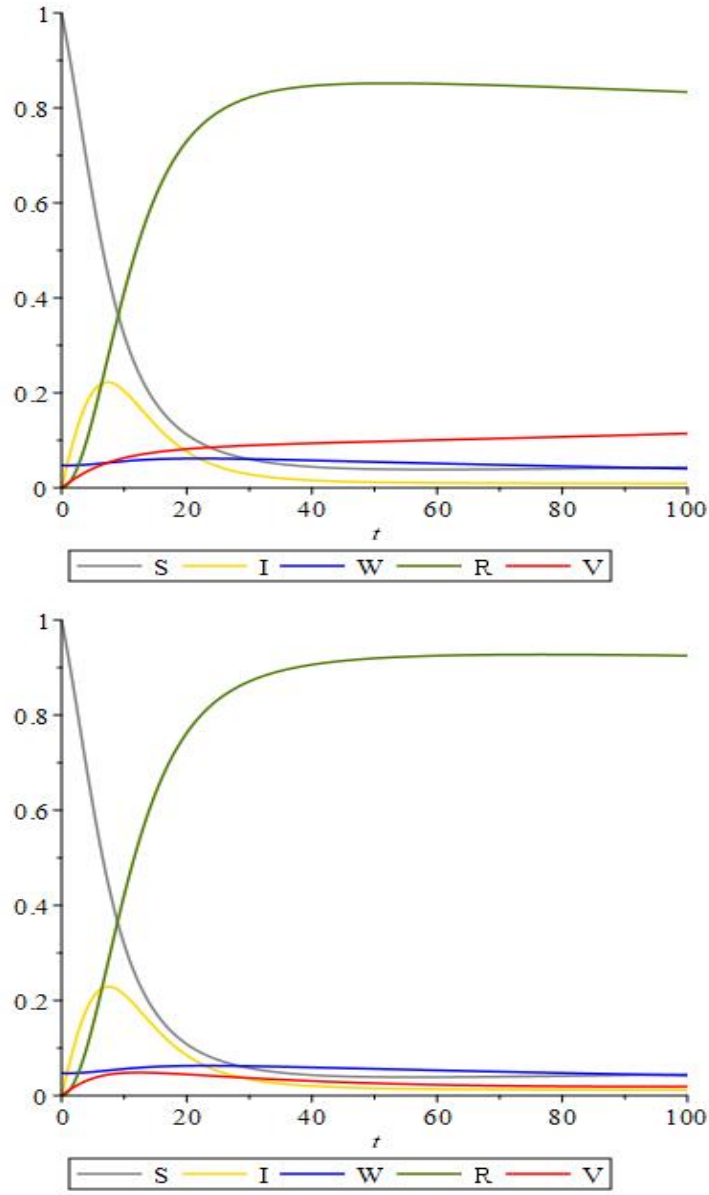


Figure 5: Perfect Vaccine, and Imperfect Vaccine ($v=0.01$)

At 1% of the population vaccinated per day, the perfect vaccine model shows considerable change in the populations, with a steady increase in the vaccinated population and a corresponding steady decrease in the susceptible and recovered populations. However, in the imperfect vaccine model, the susceptible and recovered populations show little to no change again. The vaccinated population also gets very close to 0.

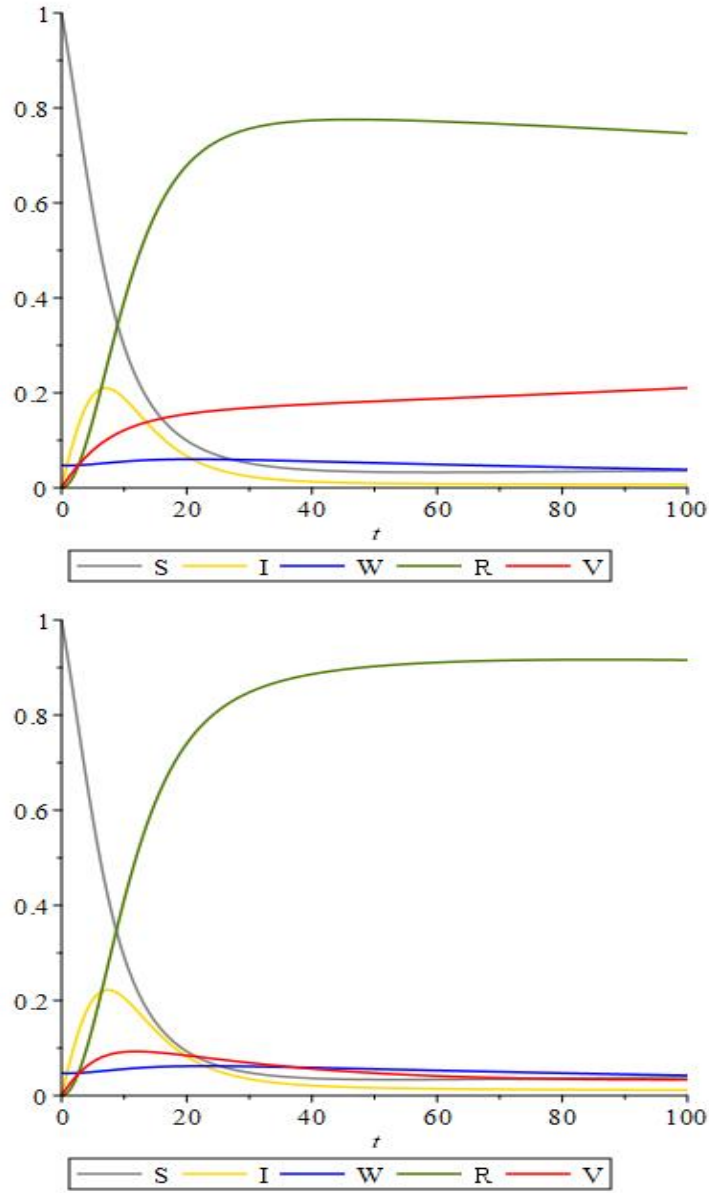


Figure 6: Perfect Vaccine, and Imperfect Vaccine ($v=0.02$)

At 2% of the population vaccinated per day, the perfect vaccine model shows a large, steady increase in the vaccinated population and a larger decrease in the recovered population. In the imperfect model, the vaccinated population does not dip as close to 0, and therefore the susceptible and recovered populations continue to steadily change. The larger slopes (the faster rates of change) in this model in comparison to the other two implies that this model will reach equilibrium more quickly.

5 Discussions

5.1 Conclusions

The graphical analysis conducted on the three models serves to demonstrate the impact of varying vaccination conditions on a population over time. The efficacy of a vaccine and vaccination rates are shown to have both short-term and long-term effects on the course of an outbreak. When given the graphs of $t=50$, although it is known that the populations will not stabilize within such a short time frame, there is still a substantial difference in the sizes of infected and recovered populations between the unvaccinated and the vaccinated populations. In the context of an actual epidemic, the smaller infected population in the vaccinated models indicates that the disease is infecting the population at a slower rate, which would give public health officials more time to react and mitigate the outbreak. On the other hand, given the long-term graphs at $t=365$, the differences between the graphs of the unvaccinated and vaccinated populations indicate the time it will take for the populations to stabilize. The graphs show that only the population vaccinated with the perfect vaccine is close to stability within a year of the outbreak. The other two populations, the population vaccinated with the imperfect vaccine and the unvaccinated population, are still considerably changing at $t=365$, and will not approach stability in the year-long time period. However, it is important to note that the susceptible population is much larger than the unvaccinated population after a year of the outbreak, providing conditions in which a second outbreak could reasonably occur.

The stability analysis of the equilibrium points of the three models shows the proportions of the population that are susceptible, infected, recovered, and vaccinated at equilibrium. As the time approaches infinity and the populations approach equilibrium, the susceptible portions of the populations are 17% for the unvaccinated population, 7.16% for the population vaccinated with the imperfect vaccine, and 2.71% for the population vaccinated with the perfect vaccine. The vaccinated proportions of the populations for the perfect versus the imperfect vaccine are 9.73% for the population vaccinated with the perfect vaccine and 2.4% for the population vaccinated with the imperfect vaccine. The proportion of the population that is susceptible indicates the likelihood of another outbreak within the population, given a change in water pathogen concentration. The proportion of the population that is vaccinated indicates how protected a population would be in the event of another outbreak. The drastic difference in susceptible and vaccinated percentages between different populations demonstrates the impact of vaccination on a population's overall health status and preparedness for an outbreak of disease.

5.2 Applications

The objective of this paper was to use modified SIWRS models to compare the spread of cholera in a theoretical population vaccinated with a perfect vaccine, a population vaccinated with an imperfect vaccine, and an unvaccinated population. Almost all of the constant parameters used were from a dataset from the 2006 Angola cholera epidemic, allowing for realistic values for factors such as transmission rates, recovery rates, and natural birth and death rates. The constant parameter for vaccine effectiveness, which was 0.59, came from the efficacy of the most widely used cholera vaccine, Vaxchora. The only non-constant parameter was vaccination rate, for which values that would be plausible for a real epidemic were used. Due to the combination of established models and previously researched values for parameters, the final models given in this paper are assumed to represent an accurate comparison between populations with different vaccination statuses.

The conclusions drawn by these models state that both vaccine efficacy and vaccination rates have an impact on the course of an outbreak in a population. An unvaccinated population affected by an outbreak of cholera would have a rapid spread of disease, take years to stabilize, and have a large proportion of the susceptible population remaining when eventually stabilized. These factors leave an unvaccinated population prone to repeated outbreaks of disease. A population vaccinated with the Vaxchora vaccine, although better off than a completely unvaccinated population, would also take around 2.2 years to stabilize and would have a considerable proportion of susceptible populations remaining once stabilized. In contrast to these, a population vaccinated with a perfect vaccine would theoretically stabilize within a year and would have a very minute proportion of susceptible population remaining after stabilization, making it unlikely that this population would suffer a second outbreak. Increased vaccination rates in a population would expedite these effects, allow for quicker stabiliza-

tion, and lower the chances of repeated outbreaks. Therefore, the results of this paper establish the importance of continuing experimentation on cholera vaccines in order to continue to improve vaccine efficacy and also promote and encourage vaccination among the general public, especially in areas with conditions that are conducive to the spread of disease. Understanding the effects of these factors and prioritizing preventative measures is crucial not only to the mitigation of disease when it spreads but also to prevent these destructive outbreaks from ever occurring in the first place.

5.3 Further Research

There are many options for further expansion of the research in this paper. For example, further research could reapply the models using parameters from other cholera epidemics, such as a similar dataset from the Broad Street cholera outbreak in 1854 in London. More data and analysis create consistency and a comprehensive understanding of the study's SIWRSV model. Other options include exploring the effects of changing variables other than vaccination rate, including plugging in other vaccine efficacy values for the imperfect vaccine model that are between 59% and 100%, and using this modified SIWRS model for other waterborne diseases. This is pragmatic because future vaccines will likely have improved efficacy, which could drastically influence the resulting stability of a population.

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